

Trial Higher School Certificate Examination

2008



Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Student Number on every page
- All questions may be attempted.
- Begin each question in a new booklet.
- All necessary working must be shown.
- Marks may be deducted for careless or poorly presented work.
- Board-approved calculators may be used.
- A list of standard integrals is included at the end of this paper.
- The mark allocated for each question is listed at the side of the question.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (15 marks) – Start a new booklet

Marks

a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\cos^2 x} dx$ 2

b) Find $\int \frac{e^{2x}}{e^x+1} dx$ 2

c) If $z = 3 - 3i$ and $w = 1 + i$, express $\frac{z^4}{w^3}$ in the form $a + ib$ 4

d) For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$

(i) Calculate the eccentricity of the ellipse. 1

(ii) Sketch the ellipse showing the co-ordinates of the foci and the equation of the directrices. 2

e) The equation $2x^3 + 5x - 3 = 0$ has roots α , β and γ .

(i) Find the polynomial equation with roots α^2 , β^2 and γ^2 . 2

(ii) Evaluate $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$ 2

Question 2 – (15 marks) – Start a new booklet

Marks

- a) (i) If α is a double zero of a polynomial $P(x)$, show that α is a single zero of $P'(x)$ 2

- (ii) Find integers m and n such that $(x+1)^2$ is a factor of $x^5 + 2x^2 + mx + n$ 3

- b) (i) Find the real numbers A, B and C such that 2

$$\frac{2x^2 + 7x - 1}{(x-2)(x^2 + x + 1)} = \frac{A}{x-2} + \frac{Bx + C}{x^2 + x + 1} \quad dx$$

- (ii) Hence find: 3

$$\int \frac{2x^2 + 7x - 1}{(x-2)(x^2 + x + 1)} dx$$

- c) Given $P(a\cos\theta, b\sin\theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, show that the equation of the tangent and the equation of the normal to the ellipse at P are given by 5

(i) $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

(ii) $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$

Prove that $OR \times OQ = a^2 e^2$ where R and Q are the x intercepts in (i) and (ii) respectively.

Question 3 – (15 marks) – Start a new booklet

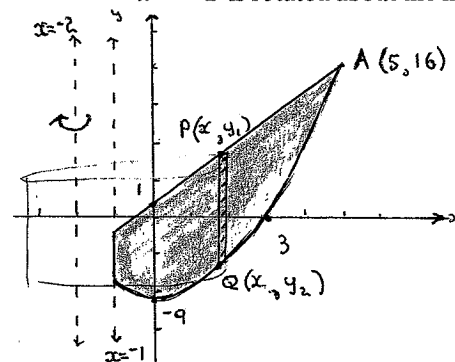
Marks

- a) Consider the function defined by $x = \theta + \frac{(\sin 2\theta)}{2}$ and $y = \theta - \frac{(\sin 2\theta)}{2}$

- (i) Show that $\frac{dy}{dx} = \tan^2 \theta$ 2

- (ii) Show that $\frac{d^2y}{dx^2} = \tan \theta \sec^4 \theta$ 2

- b) The region bounded by the curve $y = x^2 - 9$, the line $3x - y + 1 = 0$ and the line $x = -1$ is rotated about the line $x = -2$ to form a solid.



- (i) Using the method of cylindrical shells show that the volume of an elemental shell is given by 3

$$\delta V = 2\pi(x+2)(10+3x-x^2)\delta x$$

- (ii) Find the volume of the solid formed. 2

- c) The velocity, v m/s, of a particle of mass m kg moving along the x -axis is given by $v = v_0 e^{\frac{-kx}{m}}$ where v_0 is positive. Initially the particle is at the origin.

- (i) Find the displacement, x m, as a function of time. 3

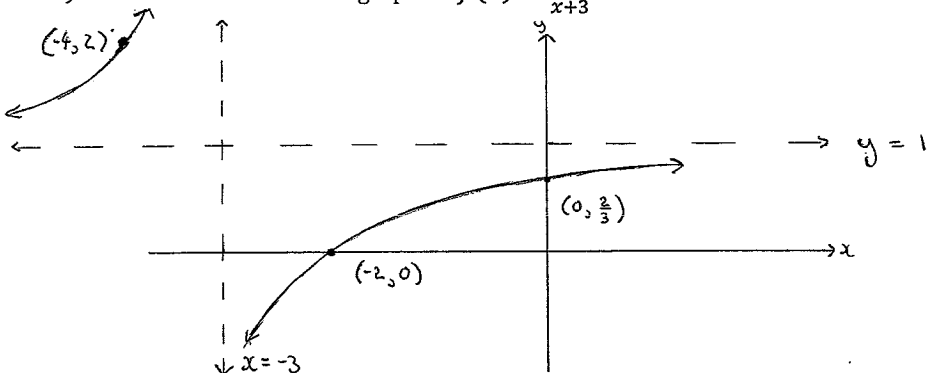
- (ii) Find the resultant force acting on this particle as a function of x 2

- (iii) Carefully describe the motion. 1

Question 4 – (15 marks) – Start a new booklet

Marks

a) Given the sketch of the graph of $f(x) = \frac{x+2}{x+3}$



Use the graph of $f(x) = \frac{x+2}{x+3}$ above to

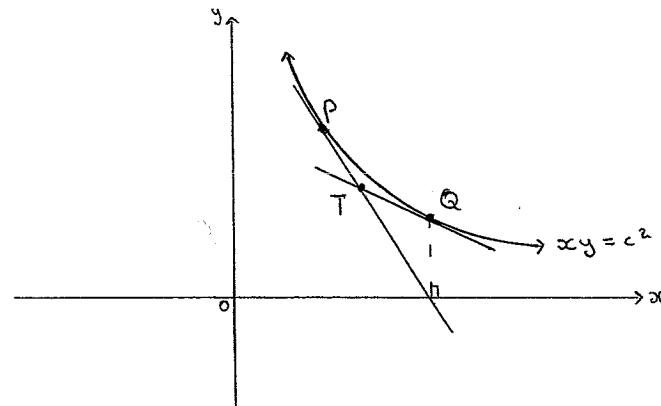
- (i) find the largest possible domain of the function $y = \sqrt{\frac{x+2}{x+3}}$ 1
- (ii) find the set of values of x for which the function $y = x - \log_e(x+3)$ is increasing. 1
- (iii) Use the graph of $f(x) = \frac{x+2}{x+3}$ above to sketch on separate axes (provided)
 - (α) the graph of $y = [f(x)]^2$ 2
 - (β) the graph of $y^2 = f(x)$ 2
 - (γ) the graph of $y = e^{f(x)}$ 2

Question 4 continued next page

Question 4 – (cont'd)

Marks

b) The distinct points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are on the same branch of the hyperbola with equation $xy = c^2$. The tangents at P and Q meet at the point T .



- (i) Show that the equation of the tangent at P is $x + p^2y = 2cp$ 2
- (ii) Show that T has co-ordinates $\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$ 2
- (iii) Let P and Q move so that the tangent at P intersects the x -axis at $(cq, 0)$. Show that the locus of T is a hyperbola and state its eccentricity. 3

Question 5 – (15 marks) – Start a new booklet

Marks

- a) A mass of m kg falls from a stationary balloon at height h metres above the ground. It experiences air resistance of mkv^2 during its fall where v is its speed in metres per second and k is a positive constant.

The equation of motion of the mass is $\ddot{x} = g - kv^2$ where g is the acceleration due to gravity.

(i) Show that $v^2 = \frac{g}{k}(1 - e^{-2kx})$

3

- (ii) Find the velocity V when the mass hits the ground.

1

(iii) Find x when $v = \frac{V}{2}$

3

- (iv) Find V if air resistance is neglected.

1

- b) For the curve $y^2 = x^2(6 + x)$

(i) By implicit differentiation show that $\frac{dy}{dx} = \frac{3x^2 + 12x}{2y}$

1

- (ii) Find any stationary points for the curve and discuss their nature.

3

- (iii) Using at least $\frac{1}{3}$ of a page, sketch the curve $y^2 = x^2(6 + x)$ showing all essential features.

2

- (iv) To calculate the area bounded by the loop, use the expression

1

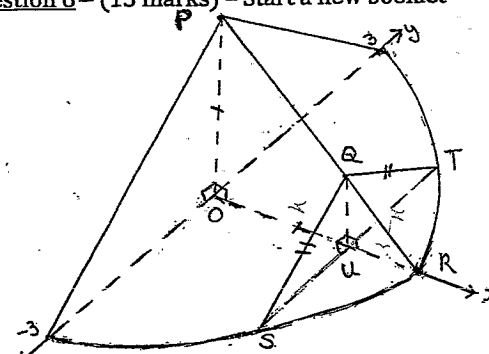
$$A = 2 \int_{-6}^0 x(6 + x)^{\frac{1}{2}} dx$$

This provides the answer $\frac{-96\sqrt{6}}{5}$. Explain why the negative sign appears on this numerical outcome.

Question 6 – (15 marks) – Start a new booklet

Marks

- a)



A solid figure has a semi-circular base of radius 3 cm. Cross sections taken perpendicular to the x -axis are isosceles triangles.

The vertical cross section containing the radius OR of the base of the solid is a right isosceles triangle ORP where $OR = OP$.

- (i) Show that the area of triangle SQT [$SQ = QT$] is given by

2

$$A = (3 - x)(9 - x^2)^{\frac{1}{2}} \text{ where } x = OU$$

- (ii) Show that the volume of this solid is $\frac{1}{4}(27\pi - 36) \text{ cm}^3$

4

- b) The polynomial $P(x)$ is defined by $P(x) = x^4 + Ax^2 + B$ where A and B are real positive numbers.

- (i) Explain why $P(x)$ has no real zeros.

3

- (ii) If two of the zeros of $P(x)$ are ib and id where b and d are real, show that $b^4 + d^4 = A^2 - 2B$

3

- c) If $I_n = \int_0^1 (1 - x^2)^n dx$ show that $I_n = \frac{2n}{2n+1} I_{n-1}$ for all positive integers $n \geq 1$

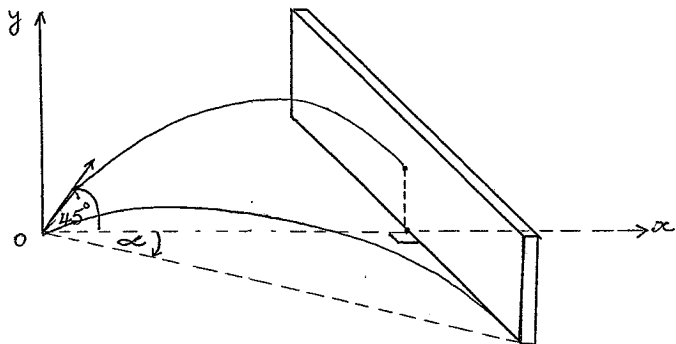
[Hint: Let $I_n = \int_0^1 (1 - x^2)(1 - x^2)^{n-1} dx$]

3

Question 7 – (15 marks) – Start a new booklet

Marks

- a) Use integration by parts to evaluate $\int_1^2 x^2 \log_e x \, dx$ 3
- b) A sprinkler is watering part of the school oval. As the water leaves the sprinkler with velocity V m/s it makes an angle θ with the ground. This angle varies continuously from 30° to 60° .
- (i) Show that the water reaches a horizontal distance R from the sprinkler where $V^2 \frac{\sqrt{3}}{2g} \leq R \leq \frac{V^2}{g}$ 4
- (ii) If this sprinkler rotates through 360° , find the area watered by the sprinkler. 1
- (iii) θ is fixed at 45° . If the sprinkler is still free to rotate through 360° and it is placed $V^2 \frac{\sqrt{3}}{2g}$ from a wall, as shown, find:
- (α) the angle of rotation, α , if the water lands exactly at the base of the wall. 2
- (β) the maximum height that the water can reach up the wall. 2



c) Solve for x : $\frac{|x|-2}{4+3x-x^2} > 0$

$-(x^2 - 3x - 4)$
 $-(x - 4)(x + 1)$

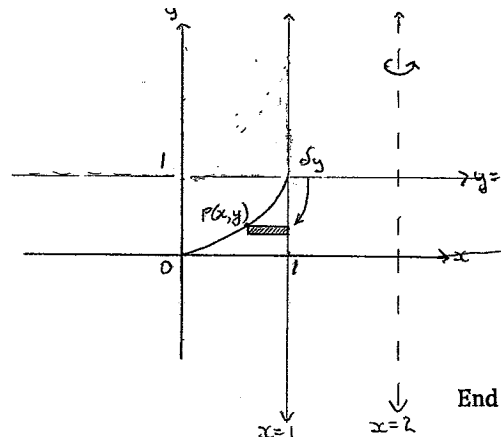
3

Question 8 – (15 marks) – Start a new booklet

Marks

- a) Show that $z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1)$ 1
- b) $z = \cos \theta + i \sin \theta$ is a root of $z^5 = 1$ where $z \neq 1$
- (i) Show that $z^2 + z + 1 + \frac{1}{z} + \frac{1}{z^2} = 0$ 2
- (ii) Let $x = z + \frac{1}{z}$. Show that $x^2 + x - 1 = 0$ 2
- (iii) Show that $z + \frac{1}{z} = 2 \cos \frac{2\pi}{5}$ or $-2 \cos \frac{\pi}{5}$ 3
- (iv) Hence show that $\cos \frac{2\pi}{5} \cdot \cos \frac{\pi}{5} = \frac{1}{4}$ 1
- (v) Find the exact value of $\cos \frac{2\pi}{5}$ 2
- c) The area bounded by $x = 1$, $y = 0$ and $y = x^2$ is rotated about the line $x = 2$.

The volume of the solid formed is to be determined by taking slices perpendicular to the axis of rotation.



- (i) Show that the area of the annulus for an elemental slice is $A = \pi[3 - 4x + x^2]$ 2
- (ii) Find the volume of the solid formed. 2

End of Paper

TRIAL HSC 2008

EXTENSION 2

SOLUTIONS

SOLUTIONS

QUESTION 1:

$$(a) \int_0^{\frac{\pi}{4}} \frac{\sin x}{1 + \cos^2 x} dx \quad \begin{array}{l} u = \cos x \\ du = -\sin x dx \end{array}$$

$$= - \int_1^0 \frac{du}{1+u^2}$$

$$= \int_0^1 \frac{du}{1+u^2}$$

$$= \left[\tan^{-1} u \right]_0^1$$

$$= \frac{\pi}{4}$$

$$(b) \int \frac{e^{2x}}{e^x + 1} dx \quad \begin{array}{l} u = e^x \\ du = e^x dx \end{array}$$

$$= \int \frac{e^x \cdot e^x dx}{e^x + 1}$$

$$= \int \frac{u}{u+1} du$$

$$= \int \left(1 - \frac{1}{u+1} \right) du$$

$$= u - \ln|u+1| + c$$

$$= e^x - \ln(e^x + 1) + c$$

$$(c) 3-3i = 3\sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)$$

$$1+i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

$$\therefore \frac{3^4}{w^3} = \frac{324 \operatorname{cis}(-\pi)}{2\sqrt{2} \operatorname{cis} \frac{3\pi}{4}}$$

$$= 81\sqrt{2} \operatorname{cis}\left(-\frac{7\pi}{4}\right)$$

$$= 81\sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

$$= 81\sqrt{2} \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i \right)$$

$$= 81 + 81i$$

$$(d) (i) a^2 = 25 \quad b^2 = 9$$

$$b^2 = a^2(1-e^2)$$

$$9 = 25(1-e^2)$$

$$\therefore 1-e^2 = \frac{9}{25}$$

$$e^2 = \frac{16}{25}$$

$$\therefore e = \frac{4}{5} \quad (e > 0)$$

$$(ii) x \text{ intercepts} = \pm 5$$

$$y \text{ intercepts} = \pm 3$$

$$\text{foci} = (\pm 4, 0)$$

$$\text{directrices: } x = \pm \frac{25}{4}$$

$$(e) 2x^3 + 5x - 3 = 0 \quad \alpha, \beta, \gamma$$

$$(i) P(\sqrt{x}) = 2(\sqrt{x})^3 + 5\sqrt{x} - 3 = 0 \quad \alpha^2, \beta^2, \gamma^2$$

$$\text{i.e. } \sqrt{x}(2x+5) = 3$$

$$x(4x+20x+25) = 9$$

$$\text{i.e. } 4x^3 + 20x^2 + 25x - 9 = 0 \text{ has roots } \alpha^2, \beta^2, \gamma^2$$

$$(ii) \Rightarrow 4\left(\frac{1}{x}\right)^3 + 20\left(\frac{1}{x}\right)^2 + 25\left(\frac{1}{x}\right) - 9 = 0 \text{ has roots } \alpha^2, \beta^2, \gamma^2$$

$$\Rightarrow 4 + 20x + 25x^2 - 9x^3 = 0$$

$$\text{i.e. } 9x^3 - 25x^2 - 20x - 4 = 0$$

$$\therefore \sum \frac{1}{x^2} = \frac{25}{9}$$

QUESTION 2:

$$(a) (i) \text{ Let } P(x) = (x-\alpha)^2 Q(x) \text{ where } Q(\alpha) \neq 0$$

$$\Rightarrow P'(x) = Q(x) \cdot 2(x-\alpha) + (x-\alpha)^2 \cdot Q'(x)$$

$$= (x-\alpha) \underbrace{[2Q(x) + (x-\alpha) \cdot Q'(x)]}_{R(x)}$$

$$\text{where } R(\alpha) = 2Q(\alpha) + 0$$

$$\neq 0 \text{ since } Q(\alpha) \neq 0$$

$$\therefore x = \alpha \text{ is a \underline{single} zero of } P'(x)$$

$$(ii) P(x) = x^5 + 2x^2 + mx + n$$

$$P(-1) = 0 \Rightarrow -1 + 2 - m + n = 0$$

$$\text{i.e. } n + m = -1 \quad \text{--- (1)}$$

$$P'(x) = 5x^4 + 4x + m$$

$$P'(-1) = 0 \Rightarrow 5 - 4 + m = 0$$

$$\therefore m = -1 \text{ sub in (1)}$$

$$\therefore n + 1 = -1$$

$$n = -2$$

$$\therefore \left. \begin{array}{l} m = -1 \\ n = -2 \end{array} \right\}$$

$$(b) (i) 2x^2 + 7x - 1 = A(x^2 + x + 1) + (Bx + C)(x - 2)$$

$$x = 2 \Rightarrow 21 = 7A$$

$$A = 3$$

$$\text{co-eff } x^2 \Rightarrow 2 = A + B$$

$$\therefore B = -1$$

$$\text{constant} \Rightarrow -1 = A - 2C$$

$$2C = 4$$

$$C = 2$$

$$\therefore \int \frac{2x^2 + 7x - 1}{(x-2)(x^2 + x + 1)} dx = \int \left(\frac{3}{x-2} + \frac{-x+2}{x^2 + x + 1} \right) dx$$

$$\begin{aligned}
 &= 3 \ln|x-2| - \frac{1}{2} \int \frac{2x+1}{x^2+x+1} + \frac{5}{2} \int \frac{dx}{(x+\frac{1}{2}) + (\frac{\sqrt{3}}{2})^2} \\
 &= 3 \ln|x-2| - \frac{1}{2} \ln|x^2+x+1| + \frac{5}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C \\
 &= 3 \ln|x-2| - \frac{1}{2} \ln(x^2+x+1) + \frac{5\sqrt{3}}{3} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C
 \end{aligned}$$

$$(c) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{d}{dx} \left(\frac{x^2}{a^2} \right) + \frac{d}{dx} \left(\frac{y^2}{b^2} \right) = 0$$

$$\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = - \frac{b^2 x}{a^2 y}$$

at $P(a \cos \theta, b \sin \theta)$

$$\begin{aligned}
 \frac{dy}{dx} &= - \frac{b^2 \cdot a \cos \theta}{a^2 \cdot b \sin \theta} \\
 &= - \frac{b \cos \theta}{a \sin \theta}
 \end{aligned}$$

(i) Tangent is:

$$y - b \sin \theta = - \frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$(a \sin \theta) y - ab \sin^2 \theta = (-b \cos \theta) x + ab \cos^2 \theta$$

$$\text{ie } (b \cos \theta) x + (a \sin \theta) y = ab$$

$$\Rightarrow \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad \text{--- (1)}$$

(ii) Normal is:

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$(b \cos \theta) y - b^2 \sin \theta \cos \theta = (a \sin \theta) x - a^2 \sin \theta \cos \theta$$

$$\therefore (a \sin \theta) x - (b \cos \theta) y = \sin \theta \cos \theta (a^2 - b^2)$$

$$\Rightarrow \frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2 \quad \text{--- (2)}$$

$$y=0 \text{ in (1)} \Rightarrow x = a \sec \theta$$

$$y=0 \text{ in (2)} \Rightarrow x = \frac{(a^2 - b^2) \cos \theta}{a}$$

$$\therefore OQ \cdot OQ' = a \sec \theta \times \frac{a^2 - b^2}{a} \cos \theta$$

$$= a^2 - b^2$$

$$b^2 = a^2 (1 - e^2)$$

$$\Rightarrow b^2 = a^2 - a^2 e^2$$

$$\therefore a^2 - b^2 = a^2 e^2$$

$$= a^2 e^2$$

QUESTION 3)

$$(a) (i) \quad x = \theta + \frac{1}{2} \sin 2\theta \quad y = \theta - \frac{1}{2} \sin 2\theta$$

$$\frac{dx}{d\theta} = 1 + \cos 2\theta \quad \frac{dy}{d\theta} = 1 - \cos 2\theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

$$= \frac{2\sin^2 \theta}{2\cos^2 \theta}$$

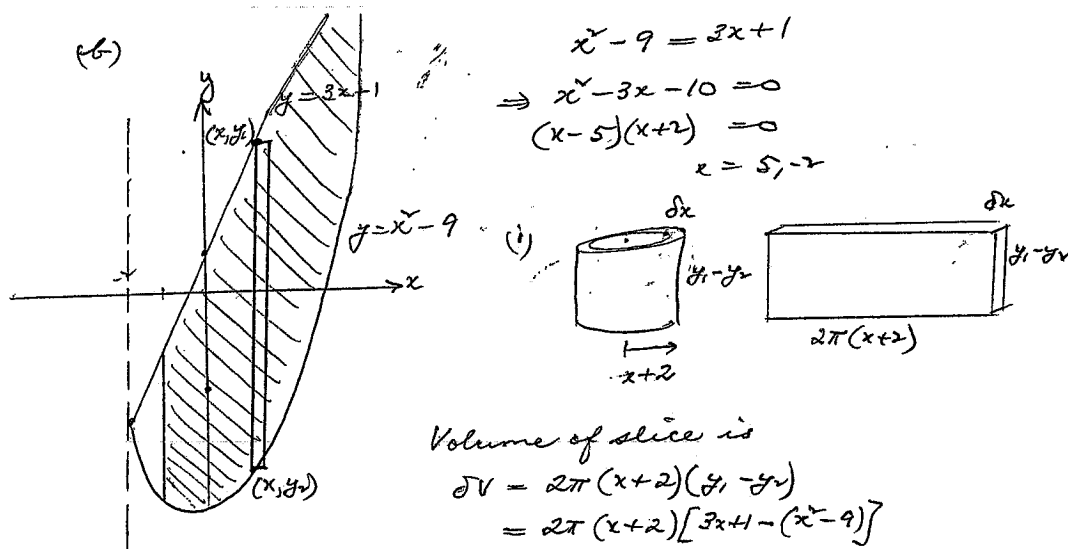
$$= \tan^2 \theta$$

$$(ii) \quad \frac{dy}{dx} = \frac{d}{dx} \left(\frac{dy}{d\theta} \right)$$

$$= \frac{d}{d\theta} \left(\frac{dy}{d\theta} \right) \cdot \frac{d\theta}{dx}$$

$$= 2 \tan \theta \sec^2 \theta \times \frac{1}{2\cos^2 \theta}$$

$$= \tan \theta \sec^4 \theta$$



Volume of solid is

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-1}^5 2\pi(x+2)(10+3x-x^2) \delta x$$

$$= 2\pi \int_{-1}^5 (x+2)(10+3x-x^2) dx$$

$$= 2\pi \int_{-1}^5 (10x + 3x^2 - x^3 + 20 + 6x - 2x^2) dx$$

$$= 2\pi \int_{-1}^5 (20 + 16x^2 + x^2 - x^3) dx$$

$$= 2\pi \left[20x + 8x^2 + \frac{x^3}{3} - \frac{x^4}{4} \right]_{-1}^5$$

$$= 2\pi \left[(100 + 200 + \frac{125}{3} - \frac{625}{4}) - (-20 + 8 - \frac{1}{3} - \frac{1}{4}) \right]$$

$$= 396\pi$$

$\therefore$ Volume is 396π units³

$$(c) \quad v = v_0 e^{-\frac{kx}{m}}, \quad v_0 > 0$$

$$(i) \quad \frac{dx}{dt} = v_0 e^{-\frac{kx}{m}}$$

$$= \frac{v_0}{e^{\frac{kx}{m}}}$$

$$\therefore \frac{dt}{dx} = \frac{e^{\frac{kx}{m}}}{v_0}$$

$$\Rightarrow t = \int \frac{e^{\frac{kx}{m}}}{v_0} dx$$

$$= \frac{1}{v_0} \cdot \frac{e^{\frac{kx}{m}}}{\left(\frac{k}{m}\right)} + C$$

$$= \frac{m}{kv_0} e^{\frac{kx}{m}} + C$$

$$\left. \begin{matrix} t=0 \\ x=0 \end{matrix} \right\} \Rightarrow 0 = \frac{m}{kv_0} + C$$

$$\therefore C = -\frac{m}{kv_0}$$

$$\therefore t = \frac{m}{kv_0} (e^{\frac{kx}{m}} - 1)$$

$$\Rightarrow \frac{kv_0 t}{m} + 1 = e^{\frac{kx}{m}}$$

$$\therefore \frac{kx}{m} = \ln\left(\frac{kv_0 t + m}{m}\right)$$

$$\therefore x = \frac{m}{k} \ln\left(\frac{kv_0 t + m}{m}\right) \quad \text{--- (1)}$$

$$(ii) \quad R = m\ddot{x}$$

$$= m v \frac{dv}{dx}$$

$$= mv_0 e^{-\frac{kx}{m}} \cdot -\frac{kv_0}{m} e^{-\frac{kx}{m}}$$

$$= -kv_0^2 e^{-\frac{2kx}{m}}$$

$$(iii) \quad \text{at } t=0, x=0, v>0 \text{ (since } v_0>0)$$

$\therefore$ Particle starts at the origin and moves to the right under a retarding force. From (1) we see that $x \rightarrow \infty$ as $t \rightarrow \infty$, also $v \rightarrow 0$ as $x \rightarrow \infty$

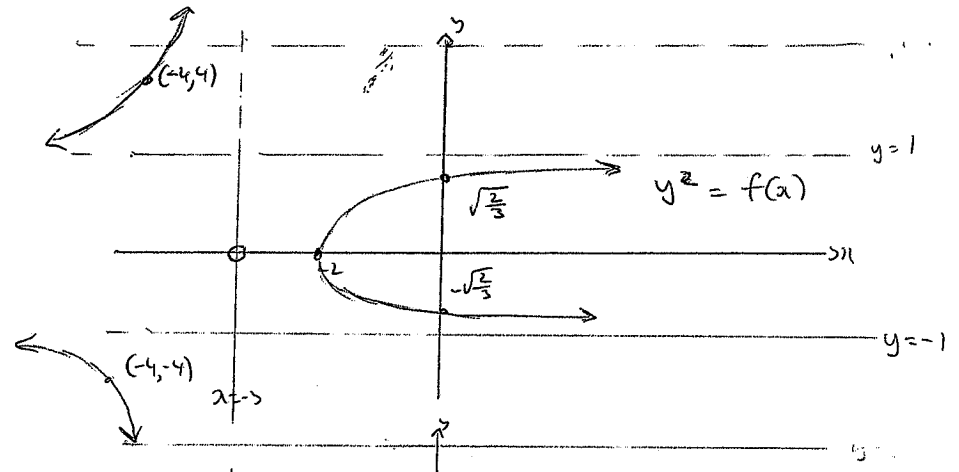
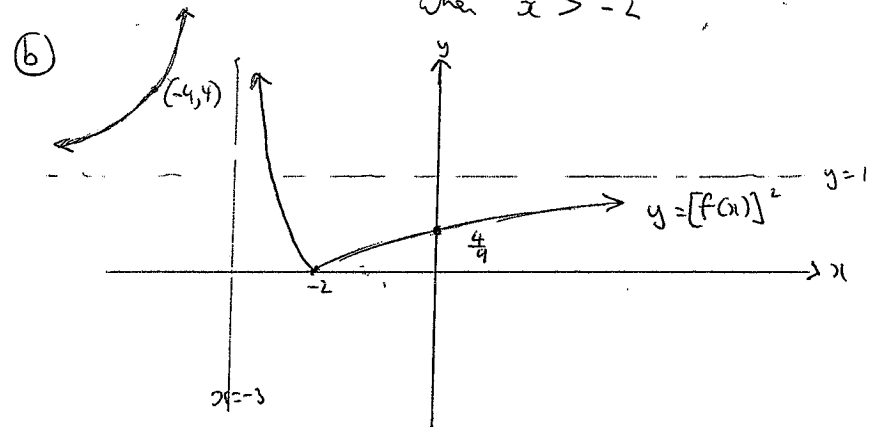
QUESTION 4:

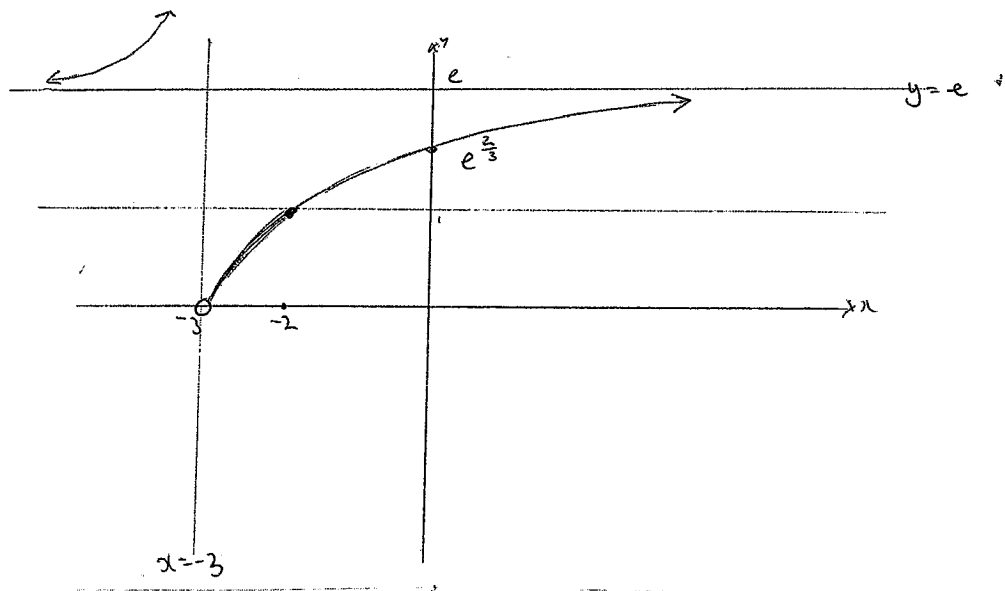
CRAMER:

(a) (i) only have $\sqrt{\quad}$ of positive values
 $\therefore$ Domain $x < -3$ and $x \geq -2$

$$(ii) \quad \text{Note: } \int \frac{x+3-1}{x+3} dx = \int 1 dx - \int \frac{dx}{x+3} \\ = x - \ln(x+3)$$

From graph area increasing when $x > -2$





(ii) Similarly, tangent at Q can be shown to be
 $x + q^2 y = 2cq$

Solving $x + p^2 y = 2cp$... I

$x + q^2 y = 2cq$... II

(I) - (II) $\Rightarrow (p^2 - q^2)y = 2c(p - q)$; (I) $(q^2 \times I) - (p^2 \times II)$
 $(p - q)(p + q)y = 2c(p - q)$ $q^2 x + p^2 q^2 y = 2cpq^2$
 $y = \frac{2c}{p+q}$ $p^2 x + p^4 q^2 y = 2cp^2 q$

$(q^2 - p^2)x = 2cpq(q - p)$
 $\therefore x = \frac{2cpq}{p+q}$

$\therefore T \left(\frac{2cpq}{p+q}, \frac{2c}{p+q} \right)$

(iii) $X = \frac{2cpq}{p+q}$... A

$y = \frac{2c}{p+q}$... B

Given $(cq, 0)$ lies on tangent at P.

Then $cq = 2cp$

$q = 2p$... C

From A $\frac{2c}{p+q} = \frac{2c}{p+2p}$

$X = \frac{4cp^2}{3p}$
 $= \frac{4cp}{3}$

From B $y = \frac{2c}{3p}$

From C, $pq = 2p^2$

Then $XY = \frac{4cp}{3} \times \frac{2c}{3p}$

$xy = \frac{8c^2}{9}$

Since $\frac{8c^2}{9}$ is a constant, the $xy = \frac{8c^2}{9}$ represent a rectangular hyperbola with eccentricity $e = \sqrt{2}$.

(i) Given $y = \frac{c^2}{x}$

$\frac{dy}{dx} = -\frac{c^2}{x^2}$

at $P(cp, \frac{c}{p})$ gradient

of tangent $m = -\frac{c^2}{(cp)^2}$
 $= -\frac{1}{p^2}$

$\therefore$ Equation of tangent

$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$
 $p^2 y - cp = -x + cp$
 $x + p^2 y = 2cp$

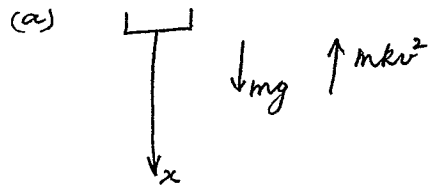
or $y + x \frac{dy}{dx} = 0$

$x \frac{dy}{dx} = -y$

$x \frac{dy}{dx} = -\frac{c^2}{x}$

$\frac{dy}{dx} = -\frac{c^2}{x^2}$

QUESTION 5:



$$(i) \quad R = m\ddot{x}$$

$$\Rightarrow m\ddot{x} = mg - mkv^2$$

$$\Rightarrow \ddot{x} = g - kv^2$$

$$(ii) \quad v \frac{dv}{dx} = g - kv^2$$

$$\Rightarrow \frac{dv}{dx} = \frac{g - kv^2}{v}$$

$$\Rightarrow \frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$\Rightarrow x = \int_0^v \frac{v}{g - kv^2} dv$$

$$= -\frac{1}{2k} [\ln(g - kv^2)]_0^v$$

$$x = -\frac{1}{2k} \left[\ln\left(\frac{g - kv^2}{g}\right) \right]$$

$$\Rightarrow -2kx = \ln\left(\frac{g - kv^2}{g}\right)$$

$$\Rightarrow e^{-2kx} = \frac{g - kv^2}{g}$$

$$ge^{-2kx} = g - kv^2$$

$$\therefore kv^2 = g(1 - e^{-2kx})$$

$$\therefore v^2 = \frac{g}{k}(1 - e^{-2kx})$$

$$(iii) \text{ when } x = h, \quad v^2 = \frac{g}{k}(1 - e^{-2kh})$$

$$\therefore V = \sqrt{\frac{g}{k}(1 - e^{-2kh})}$$

$$(iv) \text{ If } v = \frac{V}{2}$$

$$v^2 = \frac{V^2}{4}$$

$$= \frac{g}{4k}(1 - e^{-2kh})$$

$$\therefore \frac{g}{k}(1 - e^{-2kx}) = \frac{g}{4k}(1 - e^{-2kh})$$

$$4 - 4e^{-2kx} = 1 - e^{-2kh}$$

$$3 + e^{-2kh} = 4e^{-2kx}$$

$$\frac{3 + e^{-2kh}}{4} = e^{-2kx}$$

$$\therefore -2kx = \ln\left(\frac{3 + e^{-2kh}}{4}\right)$$

$$\therefore x = -\frac{1}{2k} \ln\left(\frac{3 + e^{-2kh}}{4}\right)$$

(v) If air resistance is neglected

$$\ddot{x} = g$$

$$\therefore \dot{x} = gt + c \quad \left. \begin{matrix} t=0 \\ \dot{x}=0 \end{matrix} \right\} \Rightarrow c=0$$

$$x = \frac{1}{2}gt^2 + c \quad \left. \begin{matrix} t=0 \\ x=0 \end{matrix} \right\} \Rightarrow c=0$$

$$\therefore x = \frac{1}{2}gt^2$$

$$\frac{d}{dt}\left(\frac{1}{2}gt^2\right) = g$$

$$\frac{1}{2}gt^2 = gx + c \quad \left. \begin{matrix} x=0 \\ t=0 \end{matrix} \right\} \Rightarrow c=0$$

$$\therefore v^2 = 2gx$$

$$\text{at } x = h, \quad v^2 = 2gh$$

$$\therefore V = \sqrt{2gh}$$

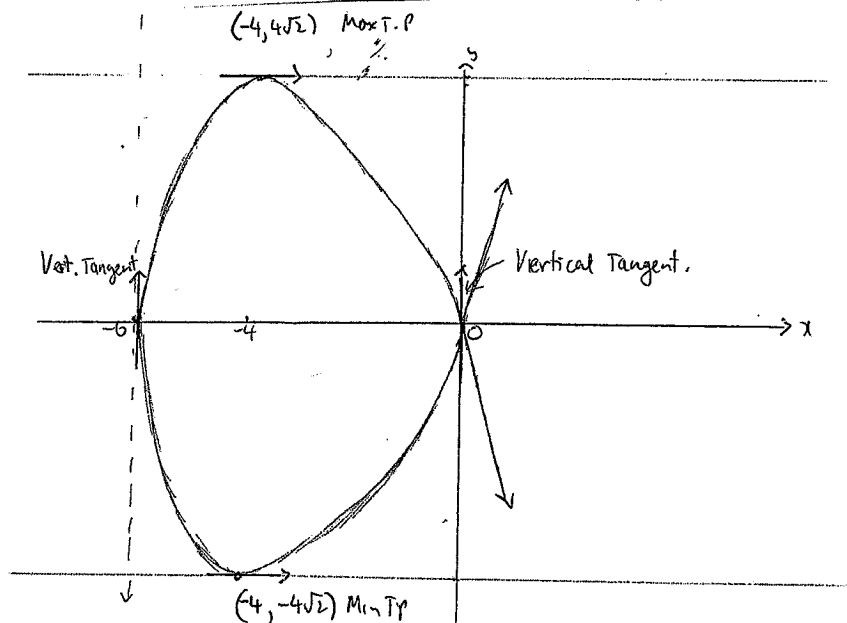
(b) (i) $2y \frac{dy}{dx} = 2x(6+x) + 1 \cdot x^2$
 $2y \frac{dy}{dx} = 12x + 3x^2$
 $\frac{dy}{dx} = \frac{12x + 3x^2}{2y}$

(ii) Stat. point $\frac{dy}{dx} = 0$
 $12x + 3x^2 = 0$
 $3x(4+x) = 0$

(a) at $x=0$ } $y=0$ gives gradient function
 $x=-6$ } as undefined. Vertical tangent at
 $(0,0)$ and $(-6,0)$

(b) at $x = -4$, $y^2 = 32$
 $\therefore y = \pm 4\sqrt{2}$

(iii)

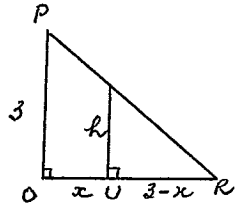


(iv) If we consider $y^2 = x^2(6+x)$
then $y = \pm \sqrt{x^2(6+x)} = \pm x(6+x)^{\frac{1}{2}}$

The area calculated is part of the loop below the x-axis.

QUESTION 6 :

(a) (i)



By similar triangles

$$\frac{h}{3} = \frac{3-x}{3}$$

$$\therefore h = 3-x$$

$$\therefore \text{Area of } \Delta PQR \text{ is } \frac{1}{2} \cdot (3-x) \cdot 2\sqrt{9-x^2} \\ = (3-x)(9-x^2)^{\frac{1}{2}}$$

(ii) Volume of solid is

$$\begin{aligned} V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^3 (3-x)(9-x^2)^{\frac{1}{2}} \delta x \\ &= \int_0^3 (3-x)(9-x^2)^{\frac{1}{2}} dx \\ &= 3 \int_0^3 \sqrt{9-x^2} dx - \int_0^3 x \sqrt{9-x^2} dx \\ &\quad \text{quadrant of circle} \quad \begin{matrix} u=9-x^2 \\ du=-2x dx \end{matrix} \\ &= 3 \times \frac{1}{4} \times \pi \cdot 9 + \frac{1}{2} \int_0^3 -2x \sqrt{9-x^2} dx \\ &= \frac{27\pi}{4} + \frac{1}{2} \int_9^0 u^{\frac{1}{2}} du \\ &= \frac{27\pi}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \left[u^{\frac{3}{2}} \right]_9^0 \\ &= \frac{27\pi}{4} + \frac{1}{3} [0 - 27] \\ &= \frac{27\pi}{4} - 9 \\ &= \frac{1}{4} (27\pi - 36) \end{aligned}$$

$$\therefore \text{Volume is } \frac{1}{4} (27\pi - 36) \text{ units}^3$$

$$(b) P(x) = x^4 + Ax^2 + B$$

$$(i) P'(x) = 4x^3 + 2Ax$$

stationary points at $P'(x) = 0$

$$\text{ie } 4x^3 + 2Ax = 0$$

$$\Rightarrow 2x(2x^2 + A) = 0$$

$$\therefore x = 0 \text{ since } 2x^2 + A \neq 0 \quad (A > 0)$$

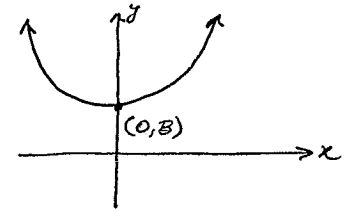
$$P(0) = B (> 0)$$

$$P''(x) = 12x^2 + 2A$$

$$> 0$$

$\therefore$ Only one stationary point at $(0, B)$

Hence sketch must be of $y = P(x)$



$\therefore P(x) = 0$ has no real zeros.

(ii) Zeros of $P(x)$ are $ib, -ib, id$ and $-id$ since all co-efficients of $P(x)$ are real

$$\sum \alpha = 0$$

$$\sum \alpha\beta = A = (ib)(-ib) + (ib)(id) + ib(-id) \\ (-ib)(id) + (-ib)(-id) + (id)(-id)$$

$$\Rightarrow b^2 - bd + bd + bd - bd + d^2 = A$$

$$\text{ie } b^2 + d^2 = A \quad \text{--- (1)}$$

$$\text{Product of roots} \Rightarrow (ib)(-ib)(id)(-id) = B$$

$$\therefore b^2 d^2 = B \quad \text{--- (2)}$$

$$\begin{aligned} \text{Then } b^4 + d^4 &= (b^2 + d^2)^2 - 2b^2 d^2 \\ &= A^2 - 2B \end{aligned}$$

$$\begin{aligned}
 (c) \quad I_n &= \int_0^1 (1-x^2)^n dx \\
 &= \int_0^1 (1-x^2)(1-x^2)^{n-1} dx \\
 &= \int_0^1 (1-x^2)^{n-1} dx - \int_0^1 x^2 (1-x^2)^{n-1} dx \\
 &= I_{n-1} - \int_0^1 \underbrace{x}_{\frac{1}{2n}} \cdot \underbrace{x(1-x^2)^{n-1}}_{\frac{d}{dx}} dx \\
 &= I_{n-1} - \left[\left[-\frac{1}{2n}(1-x^2) \cdot x \right]_0^1 - \int_0^1 -\frac{1}{2n} \cdot (1-x^2)^{n-1} dx \right] \\
 I_n &= I_{n-1} + \frac{1}{2n} \cdot I_n
 \end{aligned}$$

$$\therefore I_n + \frac{1}{2n} I_n = I_{n-1}$$

$$\left(\frac{2n+1}{2n} \right) \cdot I_n = I_{n-1}$$

$$\therefore I_n = \left(\frac{2n}{2n+1} \right) \cdot I_{n-1}$$

QUESTION 7:

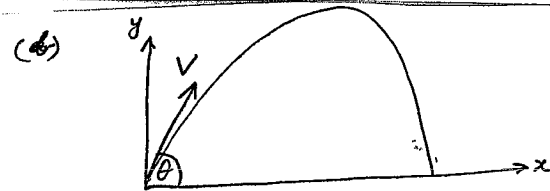
(a) let $u = \log_e x$ and $dv = x^2$

then $du = \frac{1}{x}$

$v = \frac{x^3}{3}$

$$\int u dv dx = uv - \int v du dx$$

$$\begin{aligned}
 \therefore \int_1^2 x^2 \log_e x dx &= \left[\frac{x^3}{3} \cdot \log_e x \right]_1^2 - \int_1^2 \frac{x^2}{3} dx \\
 &= \left[\frac{8}{3} \log_e 2 - 0 \right] - \left[\frac{x^3}{9} \right]_1^2 \\
 &= \frac{8}{3} \log_e 2 - \left(\frac{8}{9} - \frac{1}{9} \right) \\
 &= \frac{8}{3} \log_e 2 - \frac{7}{9}
 \end{aligned}$$



(i) $\ddot{y} = -g$

$\dot{y} = -gt + c$

$\ddot{x} = 0$

$\Rightarrow \dot{x} = V \cos \alpha$

$x = Vt \cos \alpha \quad \text{--- (1)}$

$\left. \begin{matrix} t=0 \\ y=V \sin \alpha \end{matrix} \right\} \Rightarrow c = V \sin \alpha$

$\therefore \dot{y} = V \sin \alpha - gt$

$y = Vt \sin \alpha - \frac{1}{2}gt^2 + c$

$\left. \begin{matrix} t=0 \\ y=0 \end{matrix} \right\} \Rightarrow c = 0$

$\therefore y = Vt \sin \alpha - \frac{1}{2}gt^2 \quad \text{--- (2)}$

Water hits ground at $y=0$

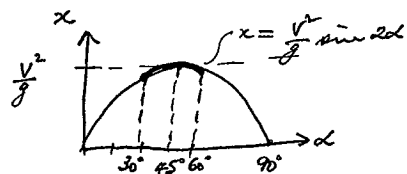
ie $Vt \sin \alpha - \frac{1}{2}gt^2 = 0$

$\frac{1}{2}t(2V \sin \alpha - gt) = 0$

$t = \frac{2V \sin \alpha}{g}$ sub in (1)

$x = V \cos \alpha \cdot \frac{2V \sin \alpha}{g}$

$= \frac{V^2 \sin 2\alpha}{g}$



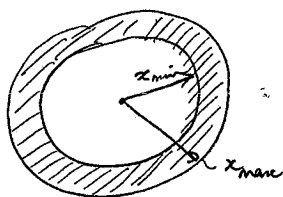
Hence $x_{\max} = \frac{V^2}{g}$ when $\alpha = 45^\circ$

$$x_{\min} = \frac{V^2}{g} \sin 60^\circ$$

$$= \frac{V^2 \sqrt{3}}{2g}$$

$$\therefore \frac{V^2 \sqrt{3}}{2g} \leq x \leq \frac{V^2}{g}$$

(ii)

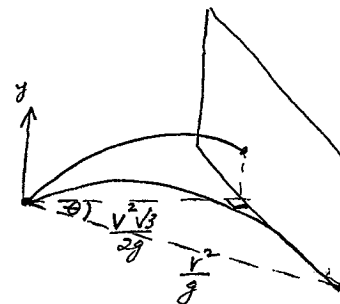


$$\text{Area watered} = \pi \left(\frac{V^2}{g} \right)^2 - \pi \left(\frac{V^2 \sqrt{3}}{2g} \right)^2$$

$$= \pi \left[\frac{V^4}{g^2} - \frac{3V^4}{4g^2} \right]$$

$$= \pi \cdot \frac{V^4}{4g^2} \text{ m}^2$$

(iii)



$$(a) \cos \theta = \frac{\frac{V^2 \sqrt{3}}{2g}}{\frac{V^2}{g}}$$

$$= \frac{V^2 \sqrt{3}}{2g} \times \frac{g}{V^2}$$

$$= \frac{\sqrt{3}}{2}$$

$$\therefore \theta = 30^\circ$$

$$(b) \text{ from (1) } x = Vt \cos \alpha$$

$$\Rightarrow Vt \cos \alpha = \frac{V^2 \sqrt{3}}{2g}$$

$$\therefore t = \frac{V \sqrt{3}}{2g \cos \alpha} \text{ where } \alpha = 45^\circ$$

$$\therefore t = \frac{V \sqrt{3}}{g \sqrt{2}} \text{ sub in (2)}$$

$$\therefore y = V \cdot \frac{1}{\sqrt{2}} \cdot \frac{V \sqrt{3}}{g \sqrt{2}} - \frac{g}{2} \cdot \frac{3V^2}{2g^2}$$

$$= \frac{V^2 \sqrt{3}}{2g} - \frac{3V^2}{4g}$$

$$= \frac{2V^2 \sqrt{3}}{4g} - \frac{3V^2}{4g}$$

$$= \frac{V^2}{4g} (2\sqrt{3} - 3)$$

$$(c) \frac{|x|-2}{(4-x)(1+x)} > 0$$

when

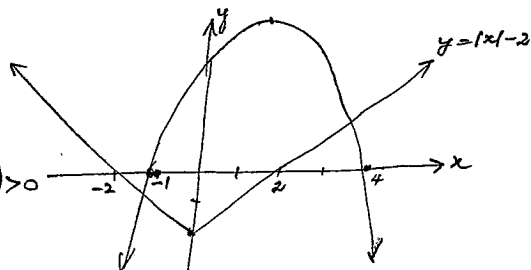
$$(i) |x|-2 > 0 \text{ and } (4-x)(1+x) > 0$$

$$\text{ie } 2 < x < 4$$

$$(ii) |x|-2 < 0 \text{ and } (4-x)(1+x) < 0$$

$$-2 < x < -1$$

$$\therefore -2 < x < -1 \text{ or } 2 < x < 4$$



QUESTION 8:

$$(a) (z-1)(z^4+z^3+z^2+z+1) = z^5+z^4+z^3+z^2+z$$

$$-z^4-z^3-z^2-z-1$$

$$= z^5-1$$

$$(b) (i) z^5=1 \Rightarrow z^5-1=0$$

$$\text{ie } (z-1)(z^4+z^3+z^2+z+1)=0$$

$$z \neq 1 \Rightarrow z^4+z^3+z^2+z+1=0$$

$$\Rightarrow z^2+z+1+\frac{1}{z}+\frac{1}{z^2}=0 \quad \text{--- (1)}$$

$$(ii) (z^2+\frac{1}{z^2})+(z+\frac{1}{z})+1=0$$

$$\Rightarrow (z+\frac{1}{z})^2-2+(z+\frac{1}{z})+1=0$$

$$x=z+\frac{1}{z} \Rightarrow x^2-2+x+1=0$$

$$\therefore x^2+x-1=0$$

$$(iii) \text{ Since } z^5=1$$

$$z = 1 \operatorname{cis} \frac{2k\pi}{5} \quad \frac{1}{z} = \operatorname{cis} \left(-\frac{2k\pi}{5}\right)$$

$$\therefore \text{Roots are } 1, z = \operatorname{cis} \frac{2\pi}{5}, z^2 = \operatorname{cis} \frac{4\pi}{5}$$

$$z^3 = \operatorname{cis} \frac{6\pi}{5}, z^4 = \operatorname{cis} \frac{8\pi}{5}$$

$$\text{Then } z + \frac{1}{z} = \operatorname{cis} \frac{2\pi k}{5} + (\operatorname{cis} \frac{2\pi k}{5})^{-1}$$

$$= \operatorname{cis} \frac{2\pi k}{5} + \operatorname{cis} \left(-\frac{2\pi k}{5}\right)$$

$$= 2 \cos \frac{2\pi k}{5}$$

$$k=1 \Rightarrow z + \frac{1}{z} = 2 \cos \frac{2\pi}{5}$$

$$k=2 \Rightarrow z + \frac{1}{z} = 2 \cos \frac{4\pi}{5} = -2 \cos \frac{\pi}{5}$$

$$k=3 \Rightarrow z + \frac{1}{z} = 2 \cos \frac{6\pi}{5} = -2 \cos \frac{\pi}{5}$$

$$k=4 \Rightarrow z + \frac{1}{z} = 2 \cos \frac{8\pi}{5} = 2 \cos \frac{2\pi}{5}$$

Hence the values of $z + \frac{1}{z}$ are $2\cos\frac{2\pi}{5}$, $-2\cos\frac{\pi}{5}$

$$\text{i.e. } x = 2\cos\frac{2\pi}{5}, -2\cos\frac{\pi}{5}$$

(iv) $\therefore x^2 + x - 1 = 0$ has roots $2\cos\frac{2\pi}{5}$, $-2\cos\frac{\pi}{5}$

Product of roots $\Rightarrow -4\cos\frac{2\pi}{5}\cos\frac{\pi}{5} = -1$

$$\therefore \cos\frac{2\pi}{5}\cos\frac{\pi}{5} = \frac{1}{4}$$

(v) $x^2 + x - 1 = 0$

$$\Rightarrow x = \frac{-1 \pm \sqrt{5}}{2} \quad \frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}$$

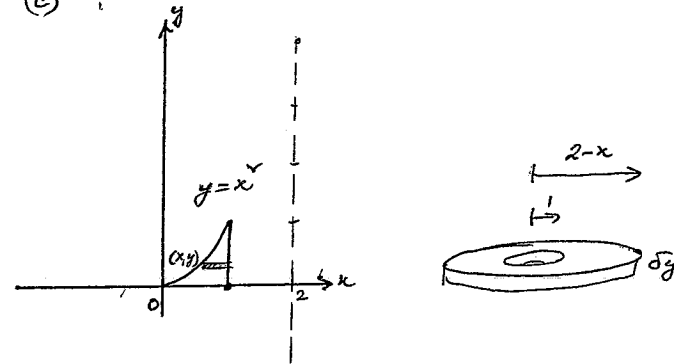
since $\cos\frac{\pi}{5} > \cos\frac{2\pi}{5}$

we see that $-2\cos\frac{\pi}{5} = \frac{-1-\sqrt{5}}{2}$

$$2\cos\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{2}$$

hence $\cos\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4}$

(c)



(i) Volume of disc is $dV = \pi(2-x)^2 dy - \pi \cdot 1^2 dy$
 $= \pi(4 - 4x + x^2 - 1) dy$
 $= \pi(3 - 4x + x^2) dy$
 $= \pi(3 - 4y^{\frac{1}{2}} + y) dy$

(ii) $\therefore$ Volume of solid is

$$V = \lim_{dy \rightarrow 0} \sum_{y=0}^1 \pi(3 - 4y^{\frac{1}{2}} + y) dy$$

$$= \pi \int_0^1 (3 - 4y^{\frac{1}{2}} + y) dy$$

$$= \pi \left[3y - 4 \cdot \frac{2}{3} y^{\frac{3}{2}} + \frac{y^2}{2} \right]_0^1$$

$$= \pi \left(3 - \frac{8}{3} + \frac{1}{2} - 0 \right)$$

$$= \frac{5\pi}{6}$$

$\therefore$ Volume is $\frac{5\pi}{6}$ units³